Project Charter

Smart Parking Platform semester project

CIS 4374 Section H — Dinh Pham

Approved August 28, 2026 at milestone M1. Recorded in this document at version 1.6.

A.1 Project Identification

Field	Value
Project name	Smart Parking Platform
Delivery organization	$oftware ¢orp.
Project sponsor	Regional parking operator group, with the municipal transportation department as co-sponsor
Project manager	Alice Nguyen
Product owner	Named by the client before requirements begin (see risk P-4, Section 8.3)
Charter date	August 28, 2026
Target delivery	December 4, 2026

A.2 Purpose and Business Case

Drivers spend a large share of every trip searching for parking, which creates congestion, wasted fuel and lost time. Operators cannot see their own occupancy in real time, so they cannot price dynamically, plan for events, or tell which assets are underused.

Both problems come from the same gap: there is no shared, live, trusted picture of which spaces are open right now.

The client previously ran three separate vendors — a consumer booking app, a gate and access control system, and a spreadsheet-based revenue process — that did not share data. The result was oversold garages and monthly reconciliation work. This project replaces that with one platform serving both sides of the transaction from a single live inventory.

A.3 Objectives and Success Criteria

ID	Objective	Measure of success
BO-1	Cut the time drivers spend searching	Median search-to-reserved under 90 seconds
BO-2	Eliminate unhonored reservations	Fewer than 0.5% of reservations fail at the gate
BO-3	Increase garage utilization	10% increase in average occupancy at participating garages
BO-4	Give operators live visibility	Occupancy dashboard reflects gate events

ID	Objective	Measure of success
		within 30 seconds
BO-5	Build driver trust in pricing	Zero difference between quoted total and charged total

A.4 High-Level Scope

In scope. Driver registration and accounts; real-time garage availability on a map; search, filter and compare; reservation create, extend and cancel; payment and refunds; digital entry pass with licence plate fallback; reservation history and receipts; notifications; the operator dashboard covering listings, capacity, pricing and live occupancy; occupancy and revenue reporting; mapping and navigation integration; administrative tools.

Out of scope for the first release. On-street and metered parking; peer-to-peer driveway rental; automated dynamic pricing; EV charging; valet; monthly and corporate permits; transit ticketing bundles.

Section 3.7 holds the full list. The out-of-scope list is what makes a 14-week delivery possible, and it is the first place scope is cut if the schedule comes under pressure.

A.5 Milestones

ID	Milestone	Date
M1	Project kickoff	Aug 28, 2026
M2	Requirements approved	Oct 2, 2026
M3	Backend feature complete	Oct 30, 2026
M4	Code complete	Nov 6, 2026
M5	UAT sign-off	Nov 20, 2026
M6	Launch	Dec 4, 2026

A.6 Budget Summary

Line	Amount
Labor (18 people, 13.10 FTE)	$540,960
Overhead and benefits at 25%	$135,240
Non-labor (infrastructure, licences, hardware, audits)	$118,300
Contingency reserve at 15%	$119,175
Budget at completion	**$913,675**

Section 9 holds the bottom-up derivation. Constraint C-1 records an unlimited theoretical development budget; this figure is therefore the performance baseline for earned value reporting, not a spending cap.

A.7 Key Stakeholders

Operators and the $oftware ¢orp. delivery teams are in the manage-closely quadrant of Figure 1 and are consulted before decisions. The city transportation department and the payment providers are kept satisfied. Drivers, mobile application users and system administrators are kept informed. Section 3.8 holds the full analysis.

A.8 High-Level Risks

ID	Risk	Score	Response
S-1	Requirements sign-off slips past October 2	9	Time-box review to one week, agree sign-off criteria up front
S-2	Garage hardware integration runs past its two weeks	6	Start vendor technical calls during requirements
T-3	Plate recognition fails in poor conditions	6	QR pass fallback plus in-app support code
P-4	Stakeholder groups want different things	6	Name one client-side product owner with final say

Seven risks score 6 or higher. Section 8 holds the full register, the triggers and the expected monetary value.

A.9 Assumptions and Constraints

Constraints. The December 4 delivery date is fixed and cannot move. The platform depends on third-party mapping, payment and gate hardware services we do not control. Payment handling must meet PCI DSS. The system must comply with city ordinance.

Assumptions. Participating garages can supply a live occupancy signal or accept our integration. Operators keep listing and pricing data current. Drivers have a smartphone with data and location access. Third-party providers remain available on commercial terms.

A.10 Project Manager Authority

The project manager may:

- Assign and reassign people across the branches in Section 5 without further approval
- Release contingency up to $25,000 per occurrence, reporting it at the next steering committee
- Re-sequence work inside a phase, provided the milestone dates in A.5 hold
- Accept or reject a deliverable against the definition of done in Section 7.7

The project manager may not change the delivery date, the budget at completion, or anything on the out-of-scope list in A.4 without steering committee approval.

A.11 Approvals

Role	Name	Date
Project sponsor	Operator group	Aug 28, 2026

Role	Name	Date
	representative	
Co-sponsor	City transportation department	Aug 28, 2026
Project manager	Alice Nguyen	Aug 28, 2026
Delivery organization	$oftware ¢orp.	Aug 28, 2026